

# MULTIMODAL CONTRASTIVE LEARNING AND DYNAMIC TEMPORAL RECONSTRUCTION FOR EEG-BASED EMOTION RECOGNITION

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## ABSTRACT

Electroencephalogram signals are crucial for emotion recognition due to high temporal resolution and non-invasiveness. However, their low signal-to-noise ratio and individual variability pose significant challenges for accurate emotion analysis. To address these issues, we propose a framework that integrates multimodal contrastive learning and dynamic temporal reconstruction (MCL-DTR) for EEG-based emotion recognition. Diverse sample views are generated by preserving rhythmic structure through frequency-domain mixing and maintaining event coherence via time-domain cropping. In addition, a dual contrastive learning strategy is introduced to mine high-quality hard negative samples. Furthermore, a dynamic temporal reconstruction module combining a transformer and deconvolution is designed to capture both long-term dependencies and transient temporal patterns in EEG signals. We adopt experiments on our proposed MCL-DTR using the SEED emotion dataset, and perform better accuracy. Experimental results suggest that contrastive learning may play a greater role in EEG signals emotion recognition tasks.

**Index Terms**— EEG signals, Emotion recognition, Contrastive learning, Temporal reconstruction, Frequency-time enhancement

## 1. INTRODUCTION

As a critical branch in human-computer interaction, affective computing focuses on the precise analysis, recognition, and understanding of human emotional states [1]. Electroencephalogram (EEG) signals, with their high temporal resolution and non-invasive advantages, can objectively capture emotion-related neural responses and have become the core data source for affective recognition [2, 3]. Despite the great potential of EEG-based emotion recognition, cross-subject generalization challenges from individual differences [4] have limited its practical

applications. Additionally, EEG signals themselves exhibit time-varying and nonlinear dynamic characteristics, and contain multi-scale information such as temporal, frequency, and spatial dimensions [5, 6]. Existing methods either insufficiently explore multi-scale structures or struggle to balance individual differences and emotional feature preservation. How to effectively extract features that are both subject-invariant and emotion-discriminative remains a key challenge in current research [7,8].

To address these issues, this study proposes an EEG emotion recognition framework that integrates time-frequency enhancement, contrastive learning, and temporal reconstruction. First, an emotion-correlated band mixing and overlapping temporal cropping strategy is designed to generate diverse training samples while maintaining emotional consistency. Second, a hard negative sample construction mechanism is introduced to form sample pairs that are highly similar yet distinguishable across emotion categories. Combined with contrastive learning, this strategy encourages the model to learn more discriminative emotional representations and sharpens category boundaries. Finally, a dynamic temporal reconstruction module is developed to capture long-range temporal dependencies and further improve the stability and discriminability of emotional features through feature reconstruction. As a result, the proposed framework significantly improves performance in cross-subject EEG emotion recognition scenarios.

## 2. PROPOSED FRAMEWORK

### 2.1. Multimodal Enhancement Strategy

In contrastive learning [9], effective data augmentation is crucial for generating high-quality positive pairs that help the encoder learn robust and invariant features. Existing methods often directly adopt techniques from computer vision or NLP without considering the spectral and temporal characteristics of EEG signals, which may introduce noise

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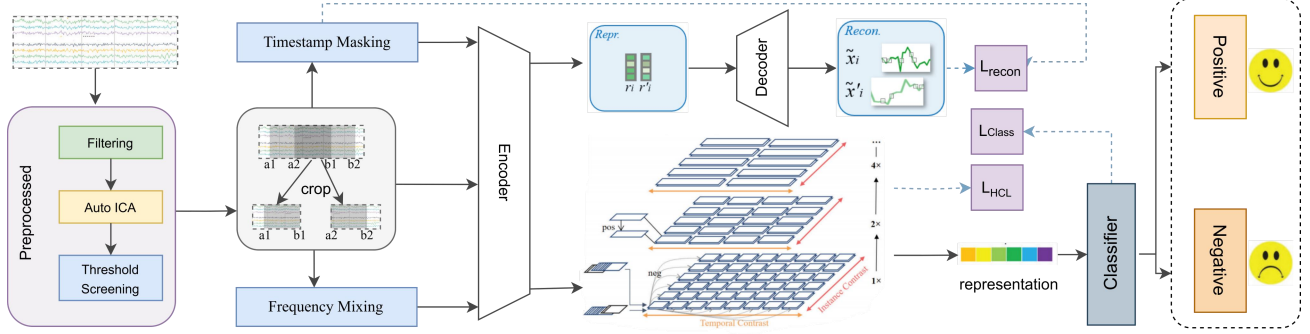


Fig.1. The network structure of the MCL-DTR framework for EEG emotion recognition.

or disrupt semantic consistency [10]. To address this, we propose a multimodal enhancement strategy that combines frequency- and time-domain operations [11] to generate diverse yet semantically consistent positive pairs.

1) **Emotion-related Frequency Mixing:** To enhance the model's ability to learn invariance in frequency-domain features, we propose a frequency mixing augmentation method. Specifically, for a randomly selected pair of training samples  $x_i$  and  $x_k$ , a specific frequency component of  $x_i$  is replaced with the corresponding frequency component from  $x_k$ . The modified signal is then reconstructed into the time domain via inverse Fast Fourier Transform. The augmented sample generated by this operation and the original sample form a positive pair. This operation effectively increases the diversity of training samples and strengthens the model's capability to extract features from emotion-related frequency bands.

2) **Overlapping Temporal Cropping:** To preserve the temporal structure of transient features, we employ a temporal cropping strategy. For each input segment  $x_i \in R^{T \times C}$  (where  $T$  is the number of time points and  $C$  is the number of channels), two time intervals  $[a_1, b_1]$  and  $[a_2, b_2]$  with an overlapping region are randomly generated, satisfying  $0 < a_1 \leq a_2 \leq b_1 \leq b_2 \leq T$ . The sub-sequences cropped from these two intervals are treated as a positive pair. The overlapping part  $[a_2, b_1]$  of the two cropped views must maintain consistent semantic representations, thereby guiding the model to capture discriminative features that are invariant across time.

## 2.2. Adversarial Negatives generation mechanism

Traditional methods that rely on randomly sampling natural negative samples often suffer from the redundancy of “easy negatives,” namely samples that differ substantially from positives and thus contribute only minimal gradients, leading to limited discriminative power of the learned representations [12, 13]. To address this limitation, we propose an adversarial hard negative generation strategy based on feature mixing. By incorporating temporal-level

and instance-level mixing, our approach synthesizes negatives that lie closer to the decision boundary, thereby posing greater challenges to the model. This mechanism effectively mitigates subject-specific and temporal variations in EEG data [14] and drives the model to learn more robust and discriminative representations for emotion recognition.

Temporal-level adversarial generation actively disrupts the temporal consistency of positive samples by introducing feature-dominated mixing from non-anchor timestamps, thereby producing negative samples that are semantically similar to the anchor yet fundamentally distinct in nature. Specifically, let  $i$  be the index of the input sequence sample and  $t$  be the timestamp.  $r_{i,t}$  and  $r'_{i,t}$  represent the same timestamp  $t$ , but are represented by two enhanced versions of  $x_i$ . The adversarial sequence of the  $i$ -th sequence at timestamp  $t$  can be expressed as:

$$\begin{aligned} r_{i,t}^{\text{temp}} &= \lambda_1 \cdot r_{i,t} + (1 - \lambda_1) \cdot r_{i,t'}, \\ r'_{i,t}^{\text{temp}} &= \lambda_1 \cdot r'_{i,t} + (1 - \lambda_1) \cdot r'_{i,t'}. \end{aligned} \quad (1)$$

Among them, it is randomly selected from  $\Omega$ , that is, the set of timestamps within the overlapping part of the two subsequences, and  $t' \neq t$ . The target timestamp enhancement represents negative sample pairs composed of temporal-level confrontation, used for calculating temporal dimension contrastive loss, helping the model capture the temporal discriminative features of the time series.

Instance-level adversarial generation aims to generate negative samples with similar semantics but different category labels by simulating the distribution differences among different subjects through feature mixing across subjects.

$$\begin{aligned} r_{i,t}^{\text{inst}} &= \lambda_2 \cdot r_{i,t} + (1 - \lambda_2) \cdot r_{j,t'}, \\ r'_{i,t}^{\text{inst}} &= \lambda_2 \cdot r'_{i,t} + (1 - \lambda_2) \cdot r'_{j,t'}, \end{aligned} \quad (2)$$

where  $j$  represents any other instance in batch  $\mathcal{B}$  except for the  $i$ -th sample. Here,  $\lambda_1$  and  $\lambda_2$  are mixing coefficients randomly selected for the anchor point, both satisfying  $\lambda_1, \lambda_2 \in (0, 0.5]$ . This value constraint ensures that the contribution of the anchor point in the mixed features is always smaller than that of the negative samples, thereby

guaranteeing the negative sample nature of the generated samples.

### 2.3. Dynamic temporal reconstruction module

Reconstruction enhances feature representations for classification tasks by forcing the model to capture both the temporal structure of the input signal and local transient features [15]. To address the limited ability of traditional reconstruction methods [16] to capture local transient features, we propose a dynamic temporal reconstruction module. By combining global dependency modeling with local detail reconstruction, the module enhances feature discriminability and interpretability in a self-supervised manner.

This module employs an encoder-decoder architecture. The encoder, built on a Transformer structure, effectively captures long-range dependencies in EEG signals via a self-attention mechanism. The decoder, composed of a deconvolutional neural network, gradually upsamples the masked time series fragments to reconstruct the original signal. Notably, the approach emphasizes the recovery of critical local regions; by minimizing the mean squared error over the masked areas, the model simultaneously preserves global contextual understanding while maintaining high sensitivity to local fine-grained features. This design not only strengthens the integration of spatiotemporal patterns relevant to emotions but also yields more discriminative feature representations for downstream classification tasks.

### 2.4. Joint optimization strategy

In this module, we jointly optimize feature learning and classification decision-making through a multi-task loss function. The overall loss function incorporates contrastive, reconstruction, and classification losses via balanced integration with hyperparameters.

We adopt a simple approach to inject double adversarial elements into contrastive learning and use them as hard negative samples in both temporal and instance contrastive losses. The two losses of the  $i$ -th time series at time  $t$  can be expressed as:

$$L_{temp}^{(i,t)} = -\log \frac{\exp(r_{i,t} \cdot r'_{i,t})}{\exp(r_{i,t} \cdot r'_{i,t}) + \sum_{z_{i,t'} \in N_i} \exp(r_{i,t} \cdot z_{i,t'})}, \quad (3)$$

$$L_{inst}^{(i,t)} = -\log \frac{\exp(r_{i,t} \cdot r'_{i,t})}{\exp(r_{i,t} \cdot r'_{i,t}) + \sum_{z_{j,t} \in N_j} \exp(r_{i,t} \cdot z_{j,t})}. \quad (4)$$

Here,  $N_i \triangleq Z_i \cup Z'_i \cup U_i \cup U'_i$ ,  $N_j \triangleq Z_j \cup Z'_j \cup U_j \cup U'_j$ .

In equations (3) and (4),  $Z_i \cup Z'_i = \{r_{i,t'}, r'_{i,t'} \mid t' \in \Omega \setminus t\}$  and  $Z_j \cup Z'_j = \{r_{j,t}, r'_{j,t} \mid j \in \{1, \dots, |\mathcal{B}|\} \setminus i\}$  are the original negative samples; And  $U_i \cup U'_i = \{r_{i,t'}^{\text{temp}}, r'_{i,t'}^{\text{temp}} \mid t' \in \Omega\}$  and

$U_j \cup U'_j = \{r_{j,t}^{\text{inst}}, r'_{j,t}^{\text{inst}} \mid j \in \{1, \dots, |\mathcal{B}|\}\}$  are the double antagonisms proposed through equations (1) and (2). Here,  $|\mathcal{B}|$  represents the batch size.

The two loss functions complement each other to capture income-specific features and reflect temporal variations. The loss is calculated and recorded at different semantic levels, that is, multiple Max pooling operations are performed on the input sequence. The Overall Hierarchical Contrastive Learning loss  $L_{dual}$  is the average of the  $L_{dual}$  losses computed over multiple Max pooling operations.

$$L_{dual} = \frac{1}{|\mathcal{B}|T} \sum_i \sum_t (L_{temp}^{(i,t)} + L_{inst}^{(i,t)}). \quad (5)$$

Masked autoencoder technology in self-supervised learning [17] has shown strong performance in fields like NLP (BERT) and computer vision (MAE) [18]. Its core idea is to reconstruct the original signal from partial observations. Inspired by this, we designed a reconstruction module to preserve important temporal variations, using the embedding function as an encoder and computing the MSE loss only on the masked timestamps, similar to BERT and MAE as shown in Formula (6):

$$L_{recon} = \frac{1}{2|\mathcal{B}|} \left\| \sum_i m_i \odot (\tilde{x}_i - x_i) \right\|_2^2 + \left\| m'_i \odot (\tilde{x}'_i - x'_i) \right\|_2^2, \quad (6)$$

here, we use  $m_i \in \{0, 1\}^{T \times F}$  to represent the observation mask of the  $i$ -th instance, where  $m_{i,t} = 0$  indicates that  $x_{i,t}$  has not been observed,  $m_{i,t} = 1$  indicates that  $x_{i,t}$  has been observed, and  $\tilde{x}_{i,t}$  is the generated reconstructed instance. In classification, we adopt cross-entropy as the loss function.

Its overall objective function is defined as follows:

$$L_{total} = L_{class} + \alpha L_{dual} + \beta L_{recon}. \quad (7)$$

Here,  $L_{class}$  represents the cross-entropy loss used in the supervised classification task, which is employed to optimize the model's discriminative ability on sentiment labels;  $L_{dual}$  is a dual contrast loss, including instance-level and sequence-level contrast terms, which is used to bring positive sample pairs closer and push negative sample pairs away, promoting the separability of the feature space.  $L_{recon}$  is the reconstruction loss, which calculates the reconstruction error of the model for the signal at the mask time point to ensure that the encoded features contain sufficient timing structure information.  $\alpha$  and  $\beta$  are balanced hyperparameters used to adjust the contribution of different loss terms during the training process.

## 3. EXPERIMENT AND RESULTS

The effectiveness of our framework is validated on the widely used SEED dataset [19]. The SEED dataset contains EEG signals collected from 15 Chinese subjects while they watched 15 movie clips of approximately 4 minutes each. Each subject completed data collection on three

experimental days, and the experimental labels were classified into three emotional states: positive, negative, and neutral. The original EEG data were collected using 62 electrodes of the International 10-20 system.

In the preprocessing stage of our research, after removing the noise and artifact components, the EEG fragments were segmented according to a 2 seconds time window and an overlap of 0.2 seconds was set to obtain the final EEG fragments for subsequent modeling. A time and channel hybrid masking strategy was introduced on the input sequence, randomly masking some time steps and electrode channels to enable the model to learn more discriminative context features. The decoder part reconstructs the Masked signal through MAE, and the optimization objective includes a weighted combination of classification cross-entropy loss and reconstruction mean square error loss. The downstream classifier adopts a one-dimensional convolution, pooling and fully connected layer structure to achieve the three-classification task.

The experiment was implemented using the pytorch framework. We adopt the Adam optimizer, with the learning rate set at  $5e-4$ . Set the batch size to 32 and train for 50 cycles in each cross-validation fold. The model performance is evaluated through five-fold cross-validation to ensure the stability and generalization of the results. All models were trained on NVIDIA GPU and an early stop mechanism was adopted to avoid overfitting.

**Table 1:** Comparison of mean accuracies (%) on the SEED dataset

| Methods        | ACC (%)      |
|----------------|--------------|
| SVM[20]        | 81.19        |
| DBN[21]        | 56.82        |
| SimCLR-EEG[22] | 85.70        |
| CLISA[23]      | 86.40        |
| MSCL[24]       | 88.18        |
| SI-CLEER[25]   | 94.25        |
| MCL-DTR(ours)  | <b>95.76</b> |

The recognition accuracy of MCL-DTR is significantly better than that of the baseline model. Compared with the pure unsupervised comparison method (SimCLR-EEG, CLISA), the multi-source contrastive learning model (MSCL), and the latest framework SI-CLEER which also uses joint learning, MCL-DTR still achieves considerable performance gain. This is mainly attributed to our proposed multimodal augmentation, adversarial hard sample generation, and dynamic temporal reconstruction modules, which together learn more discriminative and robust feature representations. The overall results show that the method proposed in this study achieves stable performance on the three-class emotion classification task.

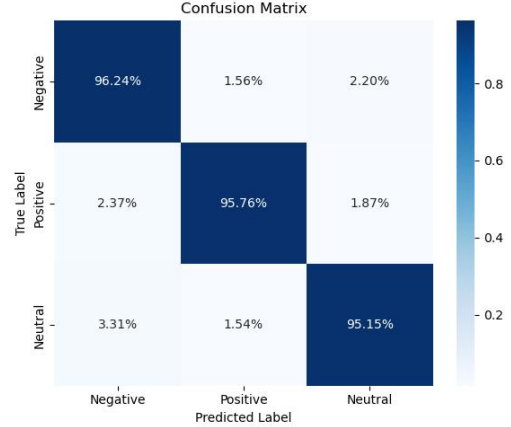
To evaluate the contribution of each module, we conducted systematic ablation studies (Table 2). Cross-band rhythm interaction and fusion proved critical for capturing emotion-related neural activity. The hard negative sampling

mechanism compressed intra-class variance and increased inter-class separation, enhancing discriminative power and robustness. The self-supervised reconstruction task complemented contrastive learning by effectively capturing temporal contextual cues beneficial for emotion recognition.

**Table 2:** Ablation study for classification accuracy on the SEED dataset. Symbol "-" indicates the following component is removed.

| Methods                          | ACC (%)      |
|----------------------------------|--------------|
| MCL-DTR(ours)                    | <b>95.76</b> |
| -Frequency domain mixing         | 93.50        |
| -Adversarial negative generation | 94.30        |
| -Dynamic time reconstruction     | 94.78        |

The category-average F1-scores were 0.9607 (negative), 0.9581 (positive), and 0.9531 (neutral), verifying the model's robust performance. Confusion matrix results indicate that the model performs well in recognizing positive and negative emotions, but shows certain confusion in the neutral category, highlighting the inherent challenge of ambiguous feature boundaries in neutral EEG signals.



**Fig.2.** Confusion matrices using the proposed MCL-DTR on the SEED dataset

## 4. CONCLUSION

In this paper, we propose an innovative framework termed MCL-DTR to address key challenges in EEG-based emotion recognition, including low signal-to-noise ratio, significant inter-subject variability, and complex spatiotemporal characteristics. The framework innovatively integrates multi-modal contrastive learning with dynamic temporal reconstruction, enabling effective extraction of emotion-discriminative features from EEG signals through an end-to-end joint optimization mechanism. Future work will focus on lightweight model design to enhance computational efficiency, and further assess the model's generalizability across broader and more culturally diverse datasets such as DEAP and DREAMER.

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